

DAILY DEMAND FORECASTING USING ARIMA

A Data Science Research Report

RabTech Academy Internship — Task 6

Research Question

Can historical daily demand patterns be used to build an effective
ARIMA forecasting model for predicting future demand?

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Abstract

This study investigates whether historical daily demand patterns can be used to develop a short-term forecasting model using the ARIMA (AutoRegressive Integrated Moving Average) methodology. The analysis uses a daily demand dataset containing 30 observations collected during June 2026. The workflow includes exploratory analysis, time-series decomposition, stationarity testing using the Augmented Dickey-Fuller test, differencing, ARIMA parameter selection, chronological train-test evaluation, and 30-day forecasting.

The stationarity analysis indicated that the original demand series was non-stationary. First-order differencing did not achieve stationarity, while second-order differencing produced a stationary series. Candidate ARIMA models were compared using the Akaike Information Criterion (AIC), and the model with the lowest AIC was selected for evaluation. Model performance was assessed using Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE).

The final model was refitted using the complete historical dataset and used to generate a 30-day forecast with 95% confidence intervals. The results demonstrate how ARIMA can be applied as a structured approach to short-term demand forecasting. However, because the dataset contains only 30 observations, the results should be considered an analytical demonstration rather than a production-ready forecasting system. Further research using a substantially larger dataset would be necessary to evaluate long-term forecasting performance and seasonal behavior.

Introduction

Demand forecasting is an important application of data science that helps organizations estimate future demand using historical observations. Accurate forecasts can support operational planning, inventory management, procurement, staffing, production scheduling, and resource allocation.

Time-series forecasting differs from ordinary predictive modeling because observations are ordered in time and may contain trends, seasonal patterns, and other temporal dependencies. Statistical forecasting methods are therefore designed to account for the structure of sequential observations.

This research focuses on the application of the AutoRegressive Integrated Moving Average (ARIMA) model to a daily demand dataset. The study follows a complete forecasting workflow beginning with data preparation and exploratory analysis, followed by time-series decomposition, stationarity testing, differencing, model selection, performance evaluation, and future forecasting.

The dataset contains 30 daily demand observations from June 2026. Because the available historical period is relatively short, the study is designed primarily as a practical demonstration of an end-to-end forecasting workflow rather than as a production forecasting system.

The central research question is: Can historical daily demand patterns be used to build an effective ARIMA forecasting model for predicting future demand?

The hypothesis is that historical daily demand contains patterns that can be modeled using ARIMA and that, after making the series stationary, the model can produce useful short-term forecasts.

Research Objectives

- Analyze the historical daily demand series.
- Examine trend, seasonality, and residual components.
- Test the stationarity of the demand series.
- Determine an appropriate differencing order.
- Compare ARIMA parameter combinations using AIC.
- Evaluate forecasting performance using MAPE and RMSE.
- Generate a 30-day future demand forecast.
- Discuss the business implications and limitations of the forecasting approach

Literature Background

Time-series forecasting has been widely used to estimate future values from historical observations. Forecasting methods are particularly useful in demand planning because organizations can use expected future demand to support inventory, procurement, production, staffing, and resource allocation decisions.

The AutoRegressive Integrated Moving Average (ARIMA) model is a classical statistical method for time-series forecasting. ARIMA combines three components: autoregression, integration through differencing, and moving-average terms. The autoregressive component uses relationships with previous observations, while the moving-average component uses previous forecast errors. Differencing is used when the original time series is not stationary.

Stationarity is an important consideration when developing an ARIMA model. A stationary time series has statistical properties that remain relatively stable over time. The Augmented Dickey-Fuller (ADF) test is commonly used to test for a unit root and assess whether a series is stationary. If a series is non-stationary, differencing can be applied before fitting an ARIMA model.

Model selection can involve comparing different combinations of ARIMA parameters. The Akaike Information Criterion (AIC) provides a measure that balances model fit and model complexity, with lower values generally indicating a more suitable model among the candidates being compared.

Forecasting performance should also be evaluated using observations that were not used to train the model. Metrics such as Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE) can be used to quantify prediction errors. Using a chronological train-test split is particularly important for time-series data because randomly mixing observations can allow information from the future to influence model training.

In this study, these established time-series concepts are applied to a daily demand dataset. The analysis focuses on stationarity, ARIMA parameter selection, out-of-sample evaluation, and short-term forecasting. The relatively small dataset also provides an opportunity to discuss the limitations of forecasting conclusions when only a short historical period is available.

Research Gap and Study context

Although ARIMA is a well-established forecasting methodology, its effectiveness depends on the characteristics and amount of available historical data. This study provides a focused application of ARIMA to a small daily demand dataset and documents the complete workflow from data preparation through future forecasting.

The study therefore emphasizes methodological transparency and reproducibility while recognizing that conclusions from a 30-observation dataset cannot be generalized broadly.

Data and Methodology

Dataset

The study uses the `daily-demand-series.csv` dataset provided for the forecasting analysis. The dataset contains 30 daily observations covering June 2026. The available variables include date, daily demand, marketing event information, and holiday information. The primary variables used in this study are date and demand.

The date variable was converted to datetime format and used as the time-series index. Observations were sorted chronologically before performing the analysis.

Data Preparation

The dataset was examined for missing values and ordered chronologically. The demand variable was treated as the primary continuous time-series outcome.

A chronological train-test split was used to preserve the temporal structure of the observations. The first 23 observations were used for model training, while the final 7 observations were reserved for testing. This approach prevents future observations from being used to train the forecasting model.

Exploratory Time-Series Analysis

Historical demand was visualized to examine changes over time. Additive time-series decomposition with a weekly period of seven days was performed to examine the trend, seasonal, and residual components of the demand series.

Stationarity Testing

The Augmented Dickey-Fuller (ADF) test was used to evaluate the stationarity of the demand series. The original series produced an ADF statistic of approximately -0.9039 and a p-value of approximately 0.7867, indicating that the original series was not stationary.

First-order differencing was then applied. The resulting series produced an ADF statistic of approximately -2.2336 and a p-value of approximately 0.1943. This did not provide sufficient evidence of stationarity.

Second-order differencing was subsequently applied. The resulting series produced an ADF statistic of approximately -15.187 with a p-value below 0.001, indicating strong evidence of stationarity. Therefore, the integration order used for the ARIMA models was set to $d = 2$.

ARIMA Modeling and Parameter Selection

Candidate ARIMA models were evaluated using combinations of autoregressive (p) and moving-average (q) parameters ranging from 0 to 2, while maintaining an integration order of $d = 2$. The Akaike Information Criterion (AIC) was used to compare the candidate models.

The model with the lowest AIC was selected for evaluation using the seven-observation test set. The selected model was subsequently refitted using the complete historical dataset before generating the final 30-day forecast.

Results

Time-Series Decomposition

The demand series was decomposed using an additive time-series decomposition with a weekly period of seven days. The decomposition separates the observed demand into trend, seasonal, and residual components. This provides a visual assessment of systematic changes and recurring weekly behavior within the available observations.

Stationarity Results

The Augmented Dickey-Fuller test indicated that the original demand series was non-stationary, with an ADF statistic of approximately -0.9039 and a p-value of 0.7867. First-order differencing produced an ADF statistic of approximately -2.2336 with a p-value of 0.1943 and therefore did not establish stationarity.

After second-order differencing, the ADF statistic was approximately -15.187 with a p-value below 0.001. This result provided strong evidence that the twice-differenced series was stationary. Consequently, an integration order of $d = 2$ was used in the ARIMA model selection process.

ARIMA Model Selection

Candidate ARIMA models with p and q values from 0 to 2 were compared while keeping $d = 2$. The Akaike Information Criterion was used to compare the candidate models, with the model having the lowest AIC selected for further evaluation.

The selected model was ARIMA(2,2,2) with an AIC of 136.3924.

Model Evaluation

The selected ARIMA(2,2,2) model was fitted using the first 23 observations and evaluated against the final 7 observations. The model produced an AIC of 136.3924. On the held-out test set, the model achieved a MAPE of 5.90% and an RMSE of 13.32 demand units.

These metrics indicate the forecasting error observed on the seven held-out observations. The MAPE expresses the average absolute error as a percentage, while RMSE represents the magnitude of prediction errors in the original demand units.

30-Day Forecast

After evaluating the selected ARIMA model on the held-out test observations, the model was refitted using the complete 30-observation demand series. The refitted model was then used to generate a 30-day forecast.

The forecast provides estimated future daily demand values together with 95% confidence intervals. The confidence intervals represent the uncertainty associated with the future predictions and generally become wider as the forecast horizon increases.

The forecast should be interpreted cautiously because the model was developed using only 30 historical observations. A longer historical dataset would provide more information about recurring patterns, seasonal effects, and longer-term changes in demand.

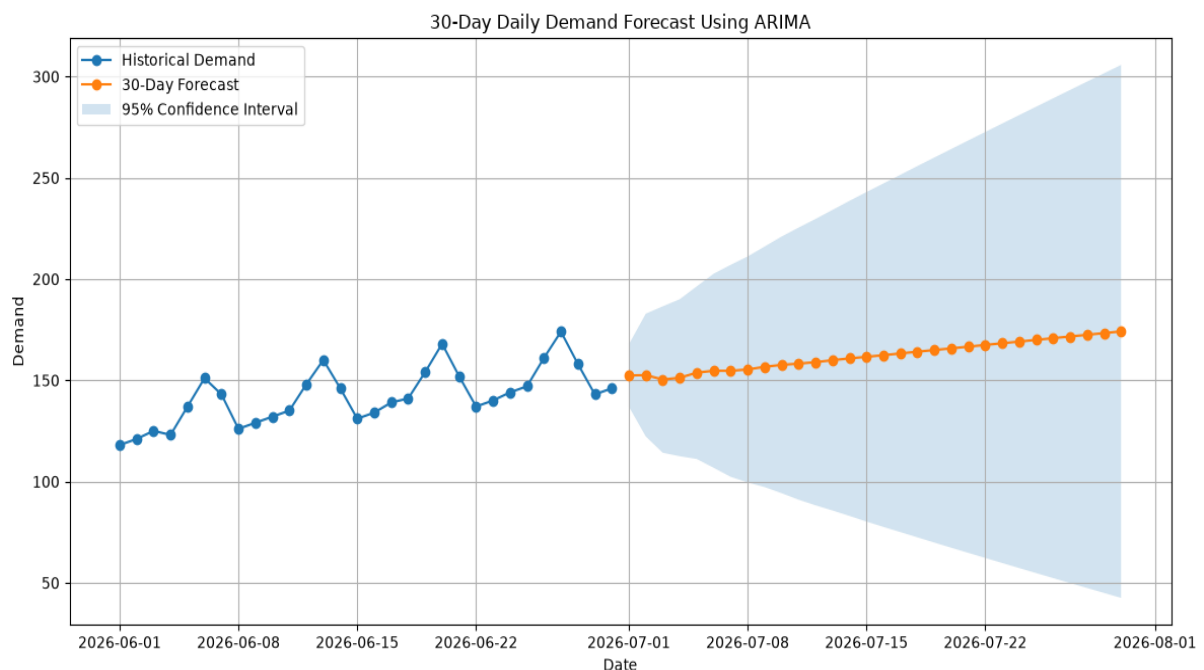


Figure 1. 30-day daily demand forecast with 95% confidence interval

Business Implications

Demand forecasting can support business decisions by providing an estimate of future demand based on historical patterns. In practical settings, forecasts can help organizations plan inventory levels, procurement activities, staffing requirements, production schedules, and resource allocation.

The ARIMA model developed in this study demonstrates how historical demand information can be converted into short-term forecasts. If similar forecasting performance were observed on a larger and more representative dataset, such forecasts could be incorporated into operational planning processes.

Forecast information could also help organizations identify periods of relatively higher or lower expected demand and adjust resources accordingly. However, forecasts should be considered alongside other relevant information, including marketing activities, holidays, external events, and changes in customer behavior.

The current study includes only 30 daily observations, so the results should not be interpreted as evidence that the model is ready for direct business deployment. A larger historical dataset and additional validation would be necessary before using the model for operational decision-making.

Limitations

This study has several limitations that should be considered when interpreting the results.

1. **Small dataset:** The analysis is based on only 30 daily observations. This is a very short historical period for time-series forecasting and limits the ability to identify reliable long-term patterns.
2. **Limited seasonal information:** Although a weekly decomposition was performed, the dataset does not contain enough historical observations to establish stable seasonal patterns across multiple weeks or months.
3. **Forecast uncertainty:** The 30-day forecast extends beyond the relatively short historical period available for model development. Consequently, forecast uncertainty may increase substantially over the longer forecast horizon.

4. Limited explanatory variables: The primary ARIMA model uses historical demand rather than explicitly incorporating variables such as marketing events or holidays as predictors.
5. Model comparison: The study focuses on ARIMA and does not provide a comprehensive comparison with alternative forecasting methods such as SARIMA, exponential smoothing, Prophet, or machine-learning models.

These limitations mean that the findings should be interpreted as a demonstration of an end-to-end forecasting methodology rather than as evidence of production-level forecasting performance.

Conclusion and Future Work

This study presented an end-to-end time-series forecasting workflow for daily demand using the ARIMA methodology. The analysis included data preparation, exploratory visualization, time-series decomposition, stationarity testing, differencing, ARIMA parameter selection, chronological train-test evaluation, and 30-day forecasting.

The Augmented Dickey-Fuller analysis showed that the original demand series was non-stationary. First-order differencing was insufficient to establish stationarity, while second-order differencing produced a stationary series. Candidate ARIMA models were subsequently compared using AIC, and the model with the lowest AIC was selected for evaluation.

The selected model was evaluated using MAPE and RMSE on the final seven observations and was then refitted using the complete historical dataset to generate a 30-day forecast with 95% confidence intervals.

The study demonstrates how statistical time-series methods can be applied to a demand forecasting problem in a reproducible workflow. However, the small number of observations limits the strength and generalizability of the findings.

Future work should use a substantially larger dataset covering several months or years. Alternative forecasting approaches such as SARIMA, exponential smoothing, Prophet, and machine-learning methods could be compared with ARIMA. Future research could also incorporate marketing events, holidays, and other external variables into forecasting models and use rolling or expanding-window validation for more robust performance assessment.

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